

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	28 September 2026
Team ID	SWTID-2026-2588
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	IT Ticket Creation	Create a new IT support ticket; Capture Caller; Capture Short Description; Capture Description
FR-2	Automatic Ticket Classification	Identify issue keywords; Automatically assign Category; Automatically assign Subcategory
FR-3	Category & Subcategory Management	Configure Category choices; Configure Subcategory choices; Implement Category–Subcategory dependency
FR-4	Flow Designer Automation	Trigger flow when a ticket is created; Check Short Description; Apply keyword-based conditions; Update ticket record automatically
FR-5	Email Notification	Send ticket confirmation email; Send notification to caller's email address; Include ticket/request details in email
FR-6	Ticket Data Management	Generate unique ticket number; Store ticket information; Maintain ticket State; Store Assigned Group and Assigned To
FR-7	Ticket Validation & Testing	Test Wi-Fi issues; Test Projector issues; Test Password/Login issues; Test Slow Computer issues
FR-8	Update Set & Deployment	Store project changes in Update Set; Complete Update Set; Export Update Set as XML for deployment

Functional Requirement Details

FR-1 – IT Ticket Creation

The system shall allow students and teachers to create an IT support ticket by providing the caller information and issue details.

FR-2 – Automatic Ticket Classification

The system shall analyze the Short Description and automatically classify the ticket based on predefined keywords.

Keyword / Issue	Category	Subcategory
Wi-Fi / Network	Network	Wi-Fi
Projector	Hardware	Projector
Password / Login	Access	Forgot Password

Slow / Hanging	Performance	Slow Computer
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FR-3 – Category & Subcategory Management

The system shall maintain a dependency between Category and Subcategory so that the relevant subcategory can be associated with the selected category.

FR-4 – Flow Designer Automation

The system shall use ServiceNow Flow Designer to trigger automation when a new ticket is created and automatically update the Category and Subcategory fields.

FR-5 – Email Notification

The system shall automatically send a confirmation email to the caller after the ticket is created.

FR-6 – Ticket Data Management

The system shall generate a unique ticket number and maintain structured ticket information including Caller, Category, Subcategory, State, Assigned Group and Assigned To.

FR-7 – Ticket Validation & Testing

The system shall be tested using different IT issue scenarios to verify that the correct category and subcategory are assigned.

FR-8 – Update Set & Deployment

The system shall store configuration changes in the Project Update Set and allow the completed Update Set to be exported for deployment.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The system should provide a simple ticket creation process and minimize the number of fields that users need to enter manually.
NFR-2	Security	Only authorized users should be able to access or modify ticket information. User and ticket data should be protected using appropriate ServiceNow access controls.
NFR-3	Reliability	The automation should consistently classify supported tickets according to the configured keyword conditions and update the correct Category and Subcategory.

NFR-4	Performance	Ticket classification and record updates should be performed automatically without unnecessary processing delays.
NFR-5	Availability	The ticket classification and notification process should be available whenever the ServiceNow platform and required services are available.
NFR-6	Scalability	The solution should support increasing numbers of IT tickets and allow additional categories, subcategories, keywords and automation rules to be added in the future.
NFR-7	Maintainability	The Flow Designer solution should be easy for ServiceNow administrators to modify and maintain without extensive custom scripting.
NFR-8	Data Integrity	Ticket information should be stored consistently, with valid Category, Subcategory, Caller and other reference-field values.
NFR-9	Extensibility	The system should allow future enhancements such as automatic assignment, SLA tracking and additional classification rules.
NFR-10	Auditability	Ticket changes and automation activities should be traceable through ServiceNow records and system logs where applicable.

Requirement Summary

Functional Requirements

Ticket Creation



Keyword Analysis



Automatic Classification



Category & Subcategory Update



Email Notification



Ticket Management

Non-Functional Requirements

Usability + Security + Reliability + Performance + Availability + Scalability + Maintainability + Data Integrity

These requirements ensure that the **Auto Ticket Classification using Flow Designer** solution is functional, reliable, maintainable, and suitable for a school IT helpdesk environment.